WORKED SOLUTION

STEP III 1987, Question 13

Mechanics • Rotating Rod with Friction

STEP III 1987, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q13

The key idea

While the particle grips the rod, rod and particle turn together as one rigid body about the hinge, so conservation of energy with the combined moment of inertia gives $\dot{\theta}^2$ at once. To find when slipping starts, we then look at the particle on its own: it moves in a circle about the hinge, so Newton's second law in the radial and transverse directions tells us exactly how much friction and normal reaction the rod must supply. Slipping begins when the friction needed reaches $\frac{1}{2}$ of the normal reaction.

Solution

The equation of motion

Model the rod as a uniform rigid rod turning about a smooth hinge O, and the particle as a point mass at distance a from O. Until the particle slides, it stays at distance a from O and turns with the rod.

The moment of inertia of the system about O is

$$I = \tfrac{4}{3}(3m)a^2 + ma^2 = 4ma^2 + ma^2 = 5ma^2,$$

using $\frac{4}{3}Ml^2$ for a uniform rod of length $2l$ about one end.

The hinge is smooth, and the friction and normal reaction between particle and rod are internal to the system, so they do no net work. When the rod has turned through θ, the rod's centre and the particle (both at distance a from O) have each fallen $a\sin\theta$. Starting from rest, conservation of energy gives

$$\tfrac{1}{2}(5ma^2)\dot{\theta}^2 = 3mg\,a\sin\theta + mg\,a\sin\theta = 4mga\sin\theta,$$

that is,

$$5a\dot{\theta}^2 = 8g\sin\theta,$$

as required. Differentiating with respect to t gives $10a\dot{\theta}\ddot{\theta} = 8g\cos\theta\,\dot{\theta}$, so (for $\dot{\theta} \neq 0$,

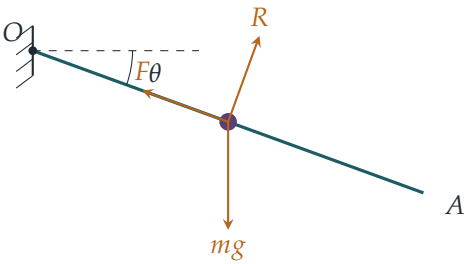
which holds once motion has begun)

$$a\ddot{\theta} = \tfrac{4}{5}g\cos\theta.$$

(The same comes from taking moments about O: $5ma^2\ddot{\theta} = 4mga\cos\theta$.)

When the particle slides

Now consider the particle alone. The forces on it are its weight mg, a normal reaction R perpendicular to the rod, and friction F along the rod. Since the particle moves on a circle of radius a about O, its acceleration has a component $a\dot{\theta}^2$ along the rod *towards* O and a component $a\ddot{\theta}$ perpendicular to the rod, in the direction of increasing θ.



We take F acting along the rod towards O, which is the direction the particle will turn out to need. Resolving along the rod, positive towards O: the weight has component $mg\sin\theta$ *away* from O, so

$$F - mg\sin\theta = ma\dot{\theta}^2 = \tfrac{8}{5}mg\sin\theta \quad\Longrightarrow\quad F = \tfrac{13}{5}mg\sin\theta.$$

Resolving perpendicular to the rod, positive in the direction of increasing θ (downwards from the rod): the weight has component $mg\cos\theta$, so

$$mg\cos\theta - R = ma\ddot{\theta} = \tfrac{4}{5}mg\cos\theta \quad\Longrightarrow\quad R = \tfrac{1}{5}mg\cos\theta.$$

Ask yourself Why does the particle need friction towards the hinge, when gravity seems to pull it down the slope?

It is not only gravity: to keep moving in a circle about O the particle needs a net force $ma\dot{\theta}^2$ towards O. Gravity's component along the rod points the other way, away from O, so friction must provide both the centripetal force and the force to cancel gravity's component. That is why F comes out positive towards O and grows like $\sin\theta$, while R is small because the rod is "falling away" beneath the particle (at $\theta = 0$ the particle has acceleration $\frac{4}{5}g$ downwards, so the rod carries only $\frac{1}{5}$ of its weight).

Both F and R are positive for $0 < \theta < \frac{\pi}{2}$, so the particle stays in contact with the rod. It stays at rest relative to the rod as long as $F \le \frac{1}{2}R$, that is,

$$\tfrac{13}{5}mg\sin\theta \le \tfrac{1}{2}\cdot\tfrac{1}{5}mg\cos\theta \iff \tan\theta \le \tfrac{1}{26}.$$

At $\theta = 0$ we have $F = 0 < \frac{1}{2}R$, and $F/R = 13\tan\theta$ increases with θ, so the particle first begins to slide (outwards, away from the hinge) when

$$\tan\theta = \tfrac{1}{26},$$

as required.

Checks

- At $\theta = 0$ the moment equation gives angular acceleration $\frac{4g}{5a}$; so the particle accelerates at $\frac{4}{5}g$, less than g, and $R = \frac{1}{5}mg > 0$ is consistent.
- Numerically stepping the rigid motion (with $g = 9.81$, $a = 0.7$) reproduced $5a\dot{\theta}^2 = 8g\sin\theta$ to 10^{-12}, and the friction and reaction found directly from the particle's computed acceleration matched $\frac{13}{5}mg\sin\theta$ and $\frac{1}{5}mg\cos\theta$. The ratio F/R first reached $\frac{1}{2}$ at $\theta \approx 0.03845$ rad, which is $\arctan\frac{1}{26} \approx 2.20°$.

Common slips

- Forgetting the centripetal term and resolving as though the particle were in equilibrium on a slope; this gives the wrong direction for friction as well as the wrong angle.
- Using $\frac{1}{3}(3m)(2a)^2$ correctly but then forgetting to add the particle's ma^2 to the moment of inertia.
- Taking $R = mg\cos\theta$, as on a fixed incline: the rod is accelerating, so the reaction is much smaller.

Beyond STEP

This question is a first taste of working in a rotating frame: viewed from the rod, the particle feels a "centrifugal" force $ma\dot{\theta}^2$ outwards and an "Euler" force $-ma\ddot{\theta}$, and these fictitious forces are exactly the terms we found. Undergraduate mechanics develops this systematically, along with Coriolis forces once the particle starts to slide. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body rotation and non-inertial frames very clearly.

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